

E88708750_1

The equation $x^4 + xy + y^3 + 4y + 1 = 0$ has no integer solutions

Size $h = 37$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + xy + y^3 + 4y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), with $(d_1, d_2) = (1, -6)$, and the fact that an odd prime p with $(-d_k/p) = -1$ divides $U_k^2 + d_k V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm forms $a^2 + db^2$

Throughout this section, d is a squarefree integer with $d \neq 0, -1$; we apply the results with $d \in \{-6, 1\}$. For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-d/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + db^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -d(bc)^2 \equiv -d \pmod{p}$, so $-d$ is a square modulo p . As $(-d/p) \neq 0$, it is a non-zero square, so $(-d/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + db^2 \neq 0$, and $v_p(a^2 + db^2)$ is even for every odd prime p with $(-d/p) = -1$.*

Proof. If $d > 0$, then clearly $a^2 + db^2 > 0$. If $d < 0$, then $-d > 1$ is squarefree, hence not a square, so $a^2 + db^2 \neq 0$. Let p be an odd prime with $(-d/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + db'^2$ by Lemma 2. Since $a^2 + db^2 = p^{2k}(a'^2 + db'^2)$, we get $v_p(a^2 + db^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{4|d|}$, or, if $d < 0$, with $n' \equiv -n \pmod{4|d|}$. Then $(-d/n') = (-d/n)$.*

Proof. Since d is squarefree, we can write $-d = \varepsilon 2^e q$ with $\varepsilon \in \{1, -1\}$, $e \in \{0, 1\}$ and q odd and positive. By multiplicativity and the reciprocity laws,

$$\left(\frac{-d}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since $4|d|$ is divisible by $4q$, and by $8q$ if $e = 1$, the claim for $n' \equiv n$ follows. If $d < 0$ and $n' \equiv -n \pmod{4|d|}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. $\square$

Lemma 5. *Let n be a non-zero integer, and let s, u and m be positive integers such that every prime factor of s divides $2|d|$, both $2s$ and $4|d|s$ divide m , u is odd, $(-d/u) = -1$, and $n \equiv su \pmod{m}$. If $d > 0$, suppose also that $n > 0$. Then there is an odd prime p with $(-d/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by $4|d|$, n' is odd and $n' \equiv u \pmod{4|d|}$. If $d > 0$, then $n' > 0$, and if $d < 0$, then $|n'| \equiv \pm u \pmod{4|d|}$; in both cases Lemma 4 gives $(-d/|n'|) = (-d/u) = -1$. By multiplicativity, the product of $(-d/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(-d/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 2d$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + xy + y^3 + 4y + 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, $d_1 = 1$, and we define

$$\begin{aligned} H_1 &= 4x^2 + x + y^2 - 4y + 4, \\ T_1 &= -x^4y + 12x^4 - 4x^2y^2 + 16x^2y + 4xy + 16x + 16y^2 + 15y + 60, \\ U_1 &= x^4 + 8x^2 + 4x + 15, \\ V_1 &= -2x^2 + 8, \\ G_1 &= -x^4 - 4x^2y + 32x^2 + 4x + 16y - 49, \\ A_1 &= -8x + 126, \\ B_1 &= -4x^3 + 63x^2 - 48x + 740, \end{aligned}$$

for the second identity, $d_2 = -6$, and we define

$$\begin{aligned} H_2 &= 2x^2 + 2xy + y^2 - 1, \\ T_2 &= 2x^5 - x^4y + 4x^4 - 4x^3y + 2x^2y^2 - 3x^2 + xy^2 + 2xy + 12x + 5y^2 \\ &\quad - y + 25, \\ U_2 &= x^4 + 2x^3 + 4x^2 - 3x, \\ V_2 &= -x^3 - 2x^2 + x + 2, \\ G_2 &= -x^4 + 2x^2y + 2x^2 + xy + 12x + 5y - 1, \\ A_2 &= -7x^2 + 11x + 62, \\ B_2 &= -7x^3 + 11x^2 + 27x + 132. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_1 T_1 - U_1^2 - V_1^2 = G_1 F, \quad (2)$$

$$H_2 T_2 - U_2^2 + 6V_2^2 = G_2 F, \quad (3)$$

$$A_1 U_1 + B_1 V_1 = 2 \cdot 5 \cdot 11 \cdot 71, \quad (4)$$

$$A_2 U_2 + B_2 V_2 = 2^3 \cdot 3 \cdot 11, \quad (5)$$

$$16H_1 = 16(y-2)^2 + (8x+1)^2 - 1. \quad (6)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_k T_k = U_k^2 + d_k V_k^2 \quad (k = 1, 2), \quad (7)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 576^2 residue pairs modulo 576, we find that (1) has exactly 288 solutions modulo 576, and that at each of them $H_1 \bmod 576$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 576$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-1/u) = -1$; here for $s = 1$, $2s = 2$ and $4s = 4$ divide 576, and for $s = 2$, every prime factor of s divides 2, and $2s = 4$ and $4s = 8$ divide 576. For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(6/u) = -1$; here for $s = 3$, every prime factor of s divides 12, and $2s = 6$ and $24s = 72$ divide 576, and for $s = 24$, every prime factor of s divides 12, and $2s = 48$ and $24s = 576$ divide 576.

Step 2: signs. *The sign of H_1 .* If $H_1 \leq 0$, then (6) gives $16(y-2)^2 \leq 1$ and $(8x+1)^2 \leq 1$, so $|y-2| \leq 0$ and $|8x+1| \leq 1$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 2)$, and it is not a solution of (1) with $H_1 \leq 0$. Hence $H_1 > 0$.

The value of H_2 . By (5), U_2 and V_2 are not both zero, so $U_2^2 - 6V_2^2 \neq 0$ by Lemma 3 with $d = -6$, and (7) gives $H_2 \neq 0$.

Step 3: primes p with $(-d_k/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(-d_k/p) = -1$, where $(d_1, d_2) = (1, -6)$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (7) and Lemma 3 show that $v_p(H_k T_k) = v_p(U_k^2 + d_k V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + d_k V_k^2$ by (7), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_k U_k + B_k V_k$. For $k = 1$, we have $R_1 = 2 \cdot 5 \cdot 11 \cdot 71$. As p is odd and $p \neq 5$ because $(-1/5) = 1$, we get $p \in \{11, 71\}$. Checking the 11^2 residue pairs modulo 11 and the 71^2 residue pairs modulo 71, we find that F , H_1 and T_1 have no common zero modulo 11 and 71, a contradiction. For $k = 2$, we have $R_2 = 2^3 \cdot 3 \cdot 11$. As p is odd and $p \neq 3$ because $(6/3) = 0$, we get $p = 11$. Checking the 11^2 residue pairs modulo 11, we find that F , H_2 and T_2 have no common zero modulo 11, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{576}$ with s and u as in Lemma 5 for $d = d_k$. By Step 2, $H_k \neq 0$, and $H_k > 0$ if $d_k > 0$. Hence Lemma 5 applies to $n = H_k$ and $d = d_k$, so there is an odd prime p with $(-d_k/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 576 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
$22 = 2 \cdot 11$	103	211	$310 = 2 \cdot 155$	391	499
31	139	$238 = 2 \cdot 119$	319	427	$526 = 2 \cdot 263$
67	$166 = 2 \cdot 83$	247	355	$454 = 2 \cdot 227$	535
$94 = 2 \cdot 47$	175	283	$382 = 2 \cdot 191$	463	571
$\mathcal{R}_2$, the residues of H_2					
$33 = 3 \cdot 11$	$168 = 24 \cdot 7$	$177 = 3 \cdot 59$	$312 = 24 \cdot 13$	$321 = 3 \cdot 107$	$465 = 3 \cdot 155$